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Alcohol and cannabis use and Co-use among U.S. sexual minority young adults: Findings from the 2019–2021 monitoring the future longitudinal panel data

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ABSTRACT

Background: Sexual minority (SM; e.g., lesbian, gay, bisexual) young adults report elevated substance use, yet few studies examine whether disparities extend to use newly-reported during young adulthood using behaviorally specific measures of daily use and co-use.

Methods: We analyzed Monitoring the Future Longitudinal Panel data (2019–2021; $N = 12{,}173$; 14.9% SM). Primary analyses used multivariable logistic regression to estimate incident-use odds during young adulthood among respondents reporting no lifetime alcohol and/or cannabis use at age 18. Outcomes included incident alcohol use, cannabis use, daily cannabis use, and past-30-day alcohol–cannabis co-use, defined as use of both substances in the past 30 days. Secondary analyses estimated prevalence and odds of past-30-day alcohol use, past-2-week binge drinking, past-30-day cannabis use, daily cannabis use, and co-use by sex and sexual identity.

Results: Gay males had the highest observed past-30-day alcohol use (80.8%) and binge drinking (42.1%), while bisexual females had significantly higher binge drinking (37.6%), past-30-day cannabis use (47.8%), daily cannabis use (19.2%), and co-use (40.9%) than heterosexual females. Lesbian females and bisexual males showed elevated prevalence, with more limited incidence elevations. Incident-use models showed elevated odds of newly-reported cannabis use and co-use among gay males and bisexual females; bisexual females also showed elevated odds of newly-reported alcohol use and daily cannabis use. Contrasts between SM subgroups were generally nonsignificant.

Conclusions: Incident-use disparities among SM young adults, particularly gay males and bisexual females, indicate that elevated substance use is not solely carried forward from adolescence. Young adulthood is therefore an important window for subgroup-specific intervention.

1. Introduction

Sexual minority (SM) individuals (e.g., lesbian, gay, bisexual) are at significantly higher risk for substance use and misuse than heterosexual peers (Krueger et al., 2020; Parent et al., 2019; Solberg et al., 2023), including elevated past-month alcohol and cannabis use in national U.S. survey data (Schilt-Solberg et al., 2025; Substance Abuse and Mental Health Services Administration, 2023). Yet research focused specifically on SM young adults remains limited (Patrick et al., 2012), with most work examining adolescents (Goldbach et al., 2014; Marshal et al., 2008; Mereish, 2019) or adults broadly (Cochran et al., 2013; Gonzales et al., 2016; Operario et al., 2015).

Young adulthood is a peak period for substance use. In 2024, 62.3% of U.S. young adults reported past-30-day alcohol use and 25.4% reported binge drinking; 29.0% reported past-30-day cannabis use and 10.8% reported daily use (Patrick et al., 2025). Alcohol-cannabis co-use increased significantly among U.S. young adults from 2002 to 2018 (McCabe et al., 2021) and is linked to more severe consequences than

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single-substance use, including substance use disorders, cognitive and mental health problems, and impaired driving (Lee et al., 2022; Tucker et al., 2023). Among SM populations, reviews indicate that pre-existing mental health difficulties, including depression, anxiety, and trauma-related symptoms, are often associated with substance use as coping (Dowling et al., 2022; Ong et al., 2026). Adverse childhood experiences are also disproportionately prevalent among SM populations and consistently linked to elevated substance use, identifying early-life trauma as one contributing pathway (McCabe et al., 2020; McCabe, Hughes, et al., 2022; McCabe, Schulenberg, et al., 2022; Ong et al., 2026).

These patterns are developmentally rooted. Transitions in education, employment, and relationships reshape peer networks, social contexts, and role obligations in ways that may constrain or facilitate substance use (Cadigan et al., 2019; Patrick, Kloska, et al., 2018; Schulenberg & Maggs, 2002; Staff et al., 2010). Habits formed during this period shape long-term health, and persistent substance use can undermine later outcomes (Lawrence et al., 2017; McCabe, Schulenberg, et al., 2022; Nelson et al., 2015; Patrick et al., 2020, 2023, Patrick, Peterson, et al., 2024; Scales et al., 2016).

Several U.S.-based studies have examined substance use among SM young adults using nationally representative data (Coulter et al., 2018; Evans-Polce et al., 2020; Fish & Exten, 2020; Liautaud et al., 2021; Schuler et al., 2018, 2019), but many rely on older cohorts (e.g., 2003–2017) and broad measures such as lifetime or past-year use. More proximal indicators, including past-30-day and daily use, better capture problematic consumption and intervention needs.

Although alcohol-cannabis co-use has been studied among young adults broadly (Lee et al., 2022), few studies focus on SM young adults, and most rely on non-representative samples (Ehlke et al., 2023; Sokolovsky et al., 2020). Recent work includes a California community cohort finding steeper increases in cannabis, but not alcohol, use among SM youth from adolescence into young adulthood (Dunbar et al., 2022), and the nationally representative National Survey on Drug Use and Health research on simultaneous alcohol and cannabis use among adult recent drinkers (Yockey et al., 2024). However, neither provides nationally representative estimates of behaviorally specific outcomes such as past-30-day and daily cannabis use among U.S. young adults, nor examines whether disparities extend to incident use during young adulthood. Daily use and co-use are especially important because they predict substance use disorders and adverse health outcomes (Connor et al., 2021; Jeffers et al., 2024; Lee et al., 2022; Tucker et al., 2023). Although prior work has examined SUDs among SM young adults (Evans-Polce et al., 2020; Fish & Exten, 2020), less is known about earlier risky patterns that may precede disorder.

Longitudinal evidence also remains limited. Foundational studies such as Add Health show that sexual identity disparities in substance use emerge by early adolescence and persist over time (Marshal et al., 2008; Needham, 2012), but most focus on adolescent initiation rather than young adulthood. This gap matters because prevalence alone cannot distinguish whether elevated SM substance use reflects patterns carried forward from adolescence or new initiation during young adulthood. This distinction has direct implications for intervention timing: earlier-onset patterns call for earlier prevention, whereas new young-adult initiation marks this period as a window of heightened risk. Prior studies identify adolescence and young adulthood as key periods for initiating alcohol and cannabis use (Marshal et al., 2008; Needham, 2012; Schulenberg & Maggs, 2002), but they do not capture outcomes such as daily cannabis use or alcohol–cannabis co-use. Important questions therefore remain about when disparities emerge and how they evolve across young adulthood.

Minority stress theory (Meyer, 2003), the predominant framework for these disparities, posits that chronic exposure to stigma, discrimination, and identity-related stressors contributes to substance use among SM individuals as a coping response. Many studies nonetheless aggregate SM individuals across sex or identity, obscuring subgroup differences: bisexual females have higher risks than both lesbian and heterosexual females (Schuler & Collins, 2020), and recent reviews identify bisexual individuals, particularly bisexual women, as a high-risk subgroup for substance use and substance use disorders (Dowling et al., 2022; Ong et al., 2026). Disaggregating by sex and sexual identity is therefore essential (Bränström et al., 2016; Bränström & Pachankis, 2018).

This study addresses these gaps using recent, nationally representative Monitoring the Future (MTF) Panel data (Patrick et al., 2025). Across sexual identity groups, we examine incident alcohol and cannabis use among respondents reporting no lifetime use at age 18 alongside current use. Comparing incidence with prevalence moves beyond documenting that disparities exist to examining whether they extend to use newly-reported during young adulthood.

2. Methods

2.1. Data and procedures

Data were collected from young adult participants in the Monitoring the Future (MTF) Longitudinal Panel study, an ongoing investigation of substance use among U.S. youth and adults (Patrick et al., 2025). An annual, nationally representative sample of 12th-grade students (base year; modal age 18) was recruited each year (Miech et al., 2024). Approximately 2450 of these students were selected for longitudinal follow-up, randomly assigned to begin one or two years post-high school (modal age 19 or 20; Patrick et al., 2025). Six biennial young adult surveys were gathered between modal ages (subsequently referred to as "age") 19 and 30.

Data from 2019 to 2021 were included because 2019 was the first year the sexual identity question appeared on all study forms (Patrick, Johnston, et al., 2024), and these are the most recent publicly available MTF data. In 2019, young adult participants were randomly assigned to the standard mailed MTF questionnaire or a web-push design (Patrick et al., 2021); starting in 2020, all young adult data collection shifted to web-push (Patrick, Schulenberg, et al., 2022). A total of 12,173 respondents participated at ages 19–30 from 2019 to 2021. The current analysis was declared exempt by the institutional review board at the University of Michigan (HUM00261717).

2.2. Measures

2.2.1. Identity

Sexual identity. To assess sexual identity, participants were asked, "Do you think of yourself as…" with response options of straight, gay or lesbian, bisexual, or other at each young adult wave.

Sex. In 2020 and earlier, sex was assessed with the item, "What is your sex?" with response options "male" and "female." In 2021, "other or prefer not to answer" was added but treated as missing due to insufficient sample size. Sex was assessed at baseline, age 18, for all respondents. Because this binary measure does not capture gender identity, it is limited in assessing how gender minority status intersects with sexual identity to shape substance use risk (see Limitations).

2.2.2. Outcomes

Past 30-day alcohol use. Respondents reported the number of occasions they used alcohol in the past 30 days (*0 occasions, 1–2, 3–5, 6–9, 10–19, 20–39, 40 or more*). Responses were dichotomized as any past-30-day alcohol use (yes/no) (Patrick et al., 2025).

Binge drinking. Binge drinking was defined by MTF as consuming five or more drinks in a row in the past two weeks, a definition that is not sex-specific. Response options were none, *once, twice, 3–5 times, 6–9 times*, or *10 or more times*, and responses were dichotomized as any binge drinking in the past two weeks (yes/no) (Patrick et al., 2025).

Past 30-day cannabis use and daily cannabis use. Respondents reported the number of occasions they used cannabis in the past 30 days (*0*

occasions, 1–2, 3–5, 6–9, 10–19, 20–39, and *40 or more occasions)*. Two dichotomous outcomes were coded: any past-30-day cannabis use (yes/no) and daily cannabis use (*≥20 occasions in the past 30 days*) (Patrick et al., 2025).

Co-use of alcohol and cannabis. Co-use was coded as a binary variable indicating use of both alcohol and cannabis in the past 30 days (yes/no) (Patrick, Kloska, et al., 2018; Venegas et al., 2022). MTF items on simultaneous use are not asked across all forms and could not be included; past-30-day co-use therefore serves as a proxy, recognizing that simultaneous use represents a riskier behavior.

2.2.3. Covariates

Time-varying covariates included age (19–30), year of data collection (2019–2021), highest degree attained by the respondent, and marital status. Time-invariant covariates included region, urbanicity, race/ethnicity, and parental education. See Table 1 for more details.

Table 1
Descriptive Statistics of Respondents in the Monitoring the Future Study (N = 12,173).

	n	%
High School Cohort		
2007–2011	4103	33.71
2012–2015	3983	32.72
2016–2020	4087	33.57
Region		
Northeast	2205	18.11
Midwest	3196	26.25
South	4090	33.60
West	2682	22.03
Urbanicity		
Urban	3965	32.57
Suburban	5882	48.32
Rural	2326	19.11
Race and Ethnicity		
White	7578	62.99
Black	976	8.11
Hispanic	1791	14.89
Asian	529	4.40
Other or Multiracial	1157	9.62
Parental Education		
At least one parent is a college graduate	6784	56.38
Neither parent is a college graduate	4998	41.54
Don't know or does not apply	251	2.09
Marital Status		
Not married	9502	80.00
Married	2375	20.00
Highest Degree Attained		
High school degree or less	5335	44.97
Associate's degree	1550	13.06
Bachelor's degree or higher	4979	41.97
Sexual Identity		
Heterosexual Males	4193	36.42
Gay Males	214	1.86
Bisexual Males	226	1.96
Other Males	64	0.56
Heterosexual Females	5604	48.68
Lesbian Females	190	1.65
Bisexual Females	856	7.44
Other Females	165	1.43
Substance Use		
Past 30-Day Alcohol Use	8181	69.57
Past 2-Week Binge Drinking	3853	34.57
Past 30-Day Cannabis Use	3545	30.07
Past 30-Day Daily Cannabis Use	1384	11.74
Past 30-Day Alcohol and Cannabis Co-Use	3028	25.97

Note. All sample sizes (n) and percentages (%) are unweighted. Cohort, region, urbanicity, race and ethnicity, and parental education were assessed at baseline (age 18). Marital status, highest degree attained, sexual identity, and substance use were assessed at follow-up waves (ages 19–30). The mean age of the sample is 24.29 years old. $N = 216$ respondents changed identity across waves and were excluded from this table. However, these respondents were still included in regression models, which allowed for a time-varying sexual identity measure.

2.3. Analysis

All analyses were conducted in R 4.5.1 using the survey package 4.0 and weighted with MTF age-specific panel weights (Patrick, Terry-McElrath, et al., 2022). Prevalence estimates for each substance use outcome were calculated by sexual identity group from 2019 to 2021. Multivariable logistic regression models were estimated using generalized estimating equations (GEE) to account for repeated observations; because respondents were surveyed biennially across a three-year window, they could contribute up to two observations. Each dichotomous substance use outcome was modeled as the dependent variable, with sexual identity and covariates entered as independent variables. All independent variables were categorical except age, 19–30, which was continuous. Survey year was categorical, with 2019, the pre-pandemic year, as the reference. Sexual identity was treated as time-varying because some respondents reported different identities across waves. As a sensitivity analysis, we excluded respondents who reported a change in sexual identity across study waves ($n = 216$). Longitudinal analyses examined incident alcohol and cannabis use from ages 19–30 among respondents reporting no lifetime use at age 18. Because respondents were not censored following first use and timing of onset was not modeled, these analyses estimate the odds of subsequently reporting use rather than the timing of initiation. For all estimates, 95% confidence intervals were calculated, and Bonferroni correction was applied for multiple comparisons. Because data collection spanned pre-pandemic and pandemic years, models adjusted for survey year, which was included as a covariate rather than examined through stratified models.

3. Results

Descriptive statistics for the overall sample ($N = 12,173$) are presented in Table 1. 36.4% of respondents identified as heterosexual males, 48.7% as heterosexual females, 1.9% as gay males, 1.7% as lesbian females, 2.0% as bisexual males, 7.4% as bisexual females, 0.6% as other males, and 1.4% as other females. Substance use prevalence among young adults ages 19–30 from 2019 to 2021 shows that 69.6% reported past 30-day alcohol use, 34.6% reported binge drinking, 30.1% reported past 30-day cannabis use, 11.7% reported daily cannabis use, and 26.0% reported co-use. We first present the prevalence of each outcome by sex and sexual identity, then examine incident (newly observed) use among respondents reporting no lifetime use at age 18 to capture use newly-reported during young adulthood, and finally report adjusted odds of current use from cross-sectional models.

3.1. Prevalence and differences of substance use outcomes by sex and sexual identity group

Prevalence estimates for each substance use outcome by sex and sexual identity group among young adults are shown in Table 2. Gay males reported the highest observed prevalence of past 30-day alcohol use (80.8%) and binge drinking (42.1%). Gay males had significantly higher prevalence of past 30-day alcohol use compared to heterosexual females (67.1%) and other females (58.7%); heterosexual males (69.2%), bisexual males (67.9%), other SM males (64.6%), lesbian females (75.7%), and bisexual females (71.5%) did not significantly differ from gay males. Gay males had a significantly higher prevalence of binge drinking than other SM females (13.7%), but not compared to heterosexual males (39.9%), bisexual males (32.0%), other SM males (34.0%), heterosexual females (29.5%), lesbian females (33.8%), or bisexual females (37.6%).

Bisexual females exhibited elevated substance use, with 47.8% reporting past 30-day cannabis use, 19.2% daily use, and 40.9% co-use of alcohol and cannabis. Bisexual females had significantly higher prevalence of cannabis use, daily cannabis use, and co-use compared to heterosexual females (23.7%, 7.0%, and 20.7%, respectively) and

Table 2
Prevalence of Substance Use by Sexual Identity Group.

	Past 30-Day Alcohol Use	Past 2-Week Binge Drinking	Past 30-Day Cannabis Use	Past 30-Day Daily Cannabis Use	Past 30-Day Alcohol and Cannabis Co-Use
	% (n) 95% CI	% (n) 95% CI	% (n) 95% CI	% (n) 95% CI	% (n) 95% CI
Males					
Heterosexual	69.22 (2908) 66.38–71.91	39.89 (1619) 37.00–42.85	29.62 (1289) 27.02–32.36	12.80 (580) 10.99–14.87	25.12 (1097) 22.69–27.72
Gay	80.82 (173) 68.63–89.03	42.11 (85) 29.29–56.09	41.42 (88) 29.07–54.96	12.39 (26) 6.11–23.51	37.51 (80) 25.62–51.12
Bisexual	67.85 (153) 54.83–78.58	32.02 (68) 21.32–45.02	37.23 (91) 26.10–49.90	19.40 (43) 11.07–31.75	33.07 (80) 22.53–45.63
Other	64.61 (37) 39.97–83.35	34.04 (16) 12.89–64.28	53.59 (28) 29.79–75.87	NA	43.33 (24) 20.88–68.90
Females					
Heterosexual	67.11 (3744) 64.70–69.44	29.48 (1531) 27.14–31.92	23.67 (1318) 21.55–25.93	7.03 (422) 5.83–8.45	20.66 (1129) 18.62–22.86
Lesbian	75.67 (142) 59.30–86.91	33.77 (60) 20.95–49.53	42.27 (71) 26.61–59.65	13.39 (24) 5.93–27.50	33.71 (58) 20.97–49.36
Bisexual	71.48 (608) 65.20–77.02	37.61 (297) 31.29–44.39	47.84 (423) 41.40–54.35	19.22 (185) 14.65–24.80	40.87 (360) 34.64–47.40
Other	58.72 (95) 43.09–72.77	13.73 (21) 6.25–27.51	40.63 (59) 26.81–56.12	18.08 (26) 9.04–32.90	36.14 (50) 22.96–51.79

Note. Cell sizes (n) are unweighted; percentages (%) and confidence intervals (CI) incorporate panel analysis weights provided by the Monitoring the Future study. Substance use behaviors were assessed at follow-up waves (ages 19–30). Cell sizes (n) represent the number of respondents in that group who indicated substance use at any follow-up wave. NA values represent prevalence corresponding to a sample size that is too small to report. Estimates are provided at the respondent-level (respondents who indicated substance use at any follow-up wave were coded as a "1"). Results for each outcome were adjusted with a Bonferroni correction to maintain a FWER of 0.05 (m = 28 for all possible pairwise comparisons). N = 216 respondents changed identity across waves and were excluded from this table. However, these respondents were included in regression models, which allowed for a time-varying sexual identity measure.

heterosexual males (29.6%, 12.8%, and 25.1%), but not compared to other SM groups. 'Other' SM males also reported high rates of past 30-day cannabis use (53.6%) and co-use (43.3%), though these estimates were not statistically significant and less stable due to small sample size.

3.2. Incident use of alcohol and cannabis by sexual identity

Among respondents reporting no lifetime use at age 18, we estimated adjusted odds of incident (newly observed) past-30-day alcohol use, binge drinking, cannabis use, daily cannabis use, and co-use during young adulthood (ages 19–30). Full results are presented in Tables 3–5. Estimates involving the 'Other' SM subgroups, and for some outcomes bisexual males, were based on small numbers of respondents and are correspondingly imprecise; contrasts with wide confidence intervals should be interpreted with caution.

3.2.1. Incident past 30-day alcohol use and binge drinking

Bisexual females had greater odds of incident past-30-day alcohol use than heterosexual males (aOR = 1.76, 95% CI: 1.05–2.94). 'Other' SM females had the lowest odds of incident alcohol use of any group, significantly lower than heterosexual males, gay males, heterosexual females, lesbian females, and bisexual females (e.g., bisexual females had 4.51 times the odds of 'Other' SM females; 95% CI: 1.69–12.01). For incident binge drinking, no stable subgroup differences emerged; the single nominally significant contrast (bisexual versus 'Other' SM females) was estimated with extreme imprecision (aOR = 27.26, 95% CI: 1.10–675.70) and we do not interpret it substantively.

3.2.2. Incident past 30-day cannabis use and daily cannabis use

Disparities in incident cannabis use were pronounced and broadly distributed across SM groups. Compared to heterosexual females, gay males (aOR = 3.39, 95% CI: 1.47–7.81), lesbian females (aOR = 2.63, 95% CI: 1.08–6.38), bisexual females (aOR = 3.22, 95% CI: 2.07–5.00), and 'Other' SM males (aOR = 6.36, 95% CI: 1.61–25.12) all had elevated odds of incident past-30-day cannabis use. Relative to heterosexual males, gay males (aOR = 2.82, 95% CI: 1.23–6.48), bisexual females (aOR = 2.68, 95% CI: 1.71–4.21), and 'Other' SM males (aOR = 5.30, 95% CI: 1.34–20.97) also showed elevated odds. For incident daily cannabis use, bisexual females (aOR = 2.95, 95% CI: 1.37–6.37) and bisexual males (aOR = 3.87, 95% CI: 1.21–12.37) had greater odds than heterosexual females.

3.2.3. Incident Co-use

Gay males and bisexual females showed elevated odds of incident alcohol–cannabis co-use relative to both heterosexual reference groups. Gay males had greater odds than heterosexual males (aOR = 3.20, 95% CI: 1.17–8.74) and heterosexual females (aOR = 2.75, 95% CI: 1.01–7.45), and bisexual females had greater odds than heterosexual males (aOR = 3.16, 95% CI: 1.33–7.50) and heterosexual females (aOR = 2.71, 95% CI: 1.15–6.41). No other subgroup contrasts reached significance.

3.3. Difference in odds of current substance use by sexual identity

Adjusted cross-sectional models of current (past-30-day) use largely paralleled the incidence findings; full results appear in Supplemental Tables 1–3. Relative to heterosexual females, gay males had elevated odds of alcohol use (aOR = 1.89, 95% CI: 1.05–3.40), and bisexual females had elevated odds of alcohol use (aOR = 1.41, 95% CI: 1.07–1.87) and cannabis use, the latter the largest disparity (aOR = 2.86, 95% CI: 2.18–3.76). Binge drinking was elevated among heterosexual males, gay males, and bisexual females, with 'Other' SM females lowest across groups. Elevated cannabis odds extended across most SM groups—gay males, lesbian females, bisexual males, bisexual females, and 'Other' SM males—and daily cannabis use was elevated among bisexual females, bisexual males, and 'Other' SM females. Co-use was elevated among bisexual females and gay males relative to both heterosexual groups, and among bisexual males and lesbian females relative to heterosexual females. Subgroup contrasts among SM groups were generally non-significant.

In sensitivity analyses excluding the 216 participants with a change in sexual identity over time, results were substantively unchanged.

4. Discussion

Using nationally representative data, we examined substance use

Table 3
Binary Logistic Regression Results Assessing the Relationship between Sexual Identity and Incident Alcohol Use from Ages 19–30.

	Incident Past 30-Day Alcohol Use (among those who reported no lifetime alcohol use at age 18) (obs. = 4845, *n* = 3811)							
Sexual Identity	aOR (95% CI)	aOR (95% CI)	aOR (95% CI)	aOR (95% CI)	aOR (95% CI)	aOR (95% CI)	aOR (95% CI)	aOR (95% CI)
Heterosexual Male	Ref.	0.51 (0.24–1.09)	0.87 (0.39–1.97)	2.35 (0.56–9.90)	0.92 (0.71–1.19)	0.58 (0.23–1.43)	0.57 (0.34–0.95)	2.57 (1.06–6.24)
Gay Male	1.97 (0.91–4.24)	Ref.	1.72 (0.59–5.05)	4.62 (0.93–23.07)	1.81 (0.85–3.86)	1.14 (0.36–3.60)	1.12 (0.46–2.71)	5.05 (1.62–15.80)
Bisexual Male	1.14 (0.51–2.58)	0.58 (0.20–1.70)	Ref.	2.69 (0.52–13.77)	1.05 (0.47–2.35)	0.66 (0.20–2.16)	0.65 (0.26–1.63)	2.94 (0.91–9.43)
Other Male	0.43 (0.10–1.79)	0.22 (0.04–1.08)	0.37 (0.07–1.91)	Ref.	0.39 (0.09–1.65)	0.25 (0.05–1.32)	0.24 (0.05–1.09)	1.09 (0.21–5.76)
Heterosexual Female	1.09 (0.84–1.41)	0.55 (0.26–1.18)	0.95 (0.43–2.13)	2.56 (0.61–10.75)	Ref.	0.63 (0.25–1.55)	0.62 (0.37–1.03)	2.79 (1.16–6.75)
Lesbian Female	1.73 (0.70–4.29)	0.88 (0.28–2.78)	1.51 (0.46–4.94)	4.06 (0.76–21.76)	1.59 (0.64–3.92)	Ref.	0.98 (0.36–2.67)	4.44 (1.31–15.12)
Bisexual Female	1.76 (1.05–2.94)	0.89 (0.37–2.16)	1.54 (0.61–3.86)	4.13 (0.92–18.52)	1.61 (0.97–2.67)	1.02 (0.37–2.75)	Ref.	4.51 (1.69–12.01)
Other Female	0.39 (0.16–0.95)	0.20 (0.06–0.62)	0.34 (0.11–1.09)	0.91 (0.17–4.82)	0.36 (0.15–0.86)	0.23 (0.07–0.77)	0.22 (0.08–0.59)	Ref.
	Incident Past Two-Week Binge Drinking (among those who reported no lifetime alcohol use at age 18) (obs. = 4605, *n* = 3652)							
Sexual Identity	aOR (95% CI)	aOR (95% CI)	aOR (95% CI)	aOR (95% CI)	aOR (95% CI)	aOR (95% CI)	aOR (95% CI)	aOR (95% CI)
Heterosexual Male	Ref.	1.01 (0.37–2.78)	1.14 (0.41–3.18)	1.68 (0.24–11.93)	1.32 (0.93–1.89)	0.97 (0.30–3.16)	0.75 (0.39–1.44)	20.48 (0.86–486.79)
Gay Male	0.99 (0.36–2.73)	Ref.	1.13 (0.28–4.62)	1.67 (0.19–14.71)	1.31 (0.48–3.61)	0.96 (0.21–4.39)	0.75 (0.24–2.35)	20.31 (0.74–557.08)
Bisexual Male	0.88 (0.31–2.45)	0.89 (0.22–3.62)	Ref.	1.47 (0.17–13.09)	1.16 (0.41–3.26)	0.85 (0.18–3.92)	0.66 (0.21–2.10)	17.98 (0.66–493.18)
Other Male	0.60 (0.08–4.23)	0.60 (0.07–5.30)	0.68 (0.08–6.02)	Ref.	0.79 (0.11–5.63)	0.58 (0.06–5.55)	0.45 (0.06–3.42)	12.19 (0.30–494.10)
Heterosexual Female	0.76 (0.53–1.08)	0.76 (0.28–2.10)	0.86 (0.31–2.41)	1.27 (0.18–9.06)	Ref.	0.73 (0.22–2.40)	0.57 (0.30–1.09)	15.46 (0.65–367.83)
Lesbian Female	1.03 (0.32–3.36)	1.04 (0.23–4.75)	1.17 (0.26–5.41)	1.73 (0.18–16.66)	1.37 (0.42–4.47)	Ref.	0.77 (0.21–2.85)	21.12 (0.68–655.09)
Bisexual Female	1.33 (0.70–2.55)	1.34 (0.42–4.24)	1.52 (0.48–4.84)	2.24 (0.29–17.12)	1.76 (0.92–3.39)	1.29 (0.35–4.75)	Ref.	27.26 (1.10–675.70)
Other Female	0.05 (0.00–1.16)	0.05 (0.00–1.35)	0.06 (0.00–1.53)	0.08 (0.00–3.32)	0.06 (0.00–1.54)	0.05 (0.00–1.47)	0.04 (0.00–0.91)	Ref.

Note. Observation counts (obs.) and sample sizes (n) are unweighted. Adjusted odds ratios (aOR) and confidence intervals (CI) incorporate wave-specific panel analysis weights provided by the Monitoring the Future study. All models control for age (time-varying), survey administration year (time-varying), region, urbanicity, race and ethnicity, parental education, highest degree attained (time-varying), and marital status (time-varying). Sexual identity was treated as time-varying. Only respondents with no lifetime alcohol use at baseline (age 18) were included in the models. Models account for repeated observations with clustering at the respondent-level. Results significant at the 0.05 level are shown in bold. Results for each outcome were adjusted with a Bonferroni correction to maintain a FWER of 0.05 (m = 28).

disparities among U.S. young adults by sexual identity. Incorporating both prevalence and incidence shows that disparities are evident not only in ongoing use but also in use newly-reported during young adulthood. Risk was concentrated among gay males and bisexual females. Bisexual females showed elevated prevalence across all five outcomes and elevated incidence for all but binge drinking; gay males showed elevated prevalence for alcohol, cannabis, and co-use, with elevated incidence for the latter two. Lesbian females and bisexual males showed elevated prevalence across several cannabis and co-use outcomes with more isolated incidence elevations, suggesting that their risk, while less pervasive, is not confined to use established before young adulthood. Because incidence models included only respondents reporting no lifetime use at age 18, they speak specifically to newly-reported use and do not quantify how much of each group's elevated prevalence reflects use carried forward from adolescence. Because onset timing was not modeled, these analyses also do not characterize developmental trajectories or establish precisely when during young adulthood disparities emerge.

Prior research shows that gay males and bisexual females engage in elevated substance use across age groups (Drabble et al., 2022; Dyar et al., 2021; McCabe et al., 2019; Schuler & Collins, 2020; Solberg et al., 2023); our findings extend this to young adulthood, a period of heightened risk (Patrick et al., 2012). Among respondents reporting no lifetime use at age 18, both groups showed elevated odds of newly-reported cannabis use and alcohol-cannabis co-use, indicating that these disparities are not attributable solely to use established in adolescence.

Alcohol use patterns varied by sex and sexual identity. Among males, heterosexual and gay men reported similarly high binge drinking prevalence and incidence, suggesting that male sex remains an important correlate in young adulthood (Patrick et al., 2025) and indicating risk for both continuing and newly-reported use; this aligns with prior work showing that SM males do not consistently exceed heterosexual males in binge drinking (Fish, 2019; Fish et al., 2017). Among females, identity differences were more pronounced: bisexual females had the highest binge drinking rates, followed by lesbian females, with heterosexual females lowest, consistent with bisexual women's heightened vulnerability. Other SM females reported the lowest prevalence and incidence. These patterns, also observed among adolescents (Kiekens et al., 2021), appear to persist into young adulthood.

Each "Other" SM subgroup comprised less than 2% of the sample and was imprecisely estimated, so results should be interpreted with caution. "Other" SM females may be less likely to frequent alcohol-centered venues that have long served as LGBTQ+ gathering spaces, instead finding community in online networks, affinity groups, or sober events (Barger et al., 2020; Bauer et al., 2020; Scroggs et al., 2023). This subgroup may also include more asexual or aromantic individuals, who report lower alcohol use (Barger et al., 2020; Scroggs et al., 2023). "Other" SM males showed the opposite pattern, with high cannabis use and co-use and elevated odds of incident cannabis use relative to both

Table 4
Binary Logistic Regression Results Assessing the Relationship between Sexual Identity and Incident Marijuana Use from Ages 19–30.

	Incident Past 30-Day Marijuana Use (among those who reported no lifetime marijuana use at age 18) (obs. = 8387, n = 6630)							
Sexual Identity	aOR (95% CI)	aOR (95% CI)	aOR (95% CI)	aOR (95% CI)	aOR (95% CI)	aOR (95% CI)	aOR (95% CI)	aOR (95% CI)
Heterosexual Male	Ref.	0.35 (0.15–0.81)	0.64 (0.31–1.36)	0.19 (0.05–0.75)	1.20 (0.89–1.61)	0.46 (0.19–1.11)	0.37 (0.24–0.59)	0.67 (0.31–1.48)
Gay Male	2.82 (1.23–6.48)	Ref.	1.82 (0.63–5.29)	0.53 (0.11–2.55)	3.39 (1.47–7.81)	1.29 (0.40–4.19)	1.05 (0.43–2.56)	1.90 (0.64–5.65)
Bisexual Male	1.55 (0.74–3.27)	0.55 (0.19–1.60)	Ref.	0.29 (0.06–1.37)	1.86 (0.89–3.91)	0.71 (0.23–2.17)	0.58 (0.26–1.30)	1.04 (0.37–2.94)
Other Male	5.30 (1.34–20.97)	1.88 (0.39–8.97)	3.41 (0.73–15.92)	Ref.	6.36 (1.61–25.12)	2.42 (0.49–12.04)	1.97 (0.48–8.10)	3.56 (0.76–16.67)
Heterosexual Female	0.83 (0.62–1.12)	0.29 (0.13–0.68)	0.54 (0.26–1.13)	0.16 (0.04–0.62)	Ref.	0.38 (0.16–0.92)	0.31 (0.20–0.48)	0.56 (0.26–1.22)
Lesbian Female	2.19 (0.90–5.31)	0.77 (0.24–2.51)	1.41 (0.46–4.31)	0.41 (0.08–2.06)	2.63 (1.08–6.38)	Ref.	0.82 (0.32–2.10)	1.47 (0.47–4.57)
Bisexual Female	2.68 (1.71–4.21)	0.95 (0.39–2.31)	1.73 (0.77–3.89)	0.51 (0.12–2.08)	3.22 (2.07–5.00)	1.23 (0.48–3.15)	Ref.	1.80 (0.78–4.20)
Other Female	1.49 (0.68–3.27)	0.53 (0.18–1.57)	0.96 (0.34–2.70)	0.28 (0.06–1.31)	1.79 (0.82–3.90)	0.68 (0.22–2.11)	0.55 (0.24–1.29)	Ref.
	Incident Past 30-Day Daily Marijuana Use (among those who reported no lifetime marijuana use at age 18) (obs. = 8387, n = 6630)							
Sexual Identity	aOR (95% CI)	aOR (95% CI)	aOR (95% CI)	aOR (95% CI)	aOR (95% CI)	aOR (95% CI)	aOR (95% CI)	aOR (95% CI)
Heterosexual Male	Ref.	0.58 (0.17–1.99)	0.42 (0.13–1.32)	0.85 (0.11–6.59)	1.62 (0.90–2.92)	0.51 (0.11–2.27)	0.55 (0.25–1.20)	0.56 (0.15–2.14)
Gay Male	1.73 (0.50–5.94)	Ref.	0.72 (0.15–3.56)	1.46 (0.14–15.18)	2.81 (0.81–9.76)	0.88 (0.13–5.77)	0.95 (0.25–3.66)	0.97 (0.17–5.52)
Bisexual Male	2.39 (0.76–7.50)	1.38 (0.28–6.77)	Ref.	2.02 (0.21–19.68)	3.87 (1.21–12.37)	1.21 (0.20–7.45)	1.31 (0.37–4.63)	1.34 (0.25–7.07)
Other Male	1.18 (0.15–9.21)	0.68 (0.07–7.10)	0.50 (0.05–4.84)	Ref.	1.92 (0.24–15.15)	0.60 (0.05–7.24)	0.65 (0.08–5.36)	0.67 (0.06–7.19)
Heterosexual Female	0.62 (0.34–1.10)	0.36 (0.10–1.24)	0.26 (0.08–0.82)	0.52 (0.07–4.11)	Ref.	0.31 (0.07–1.45)	0.34 (0.16–0.73)	0.35 (0.09–1.33)
Lesbian Female	1.97 (0.44–8.81)	1.14 (0.17–7.49)	0.83 (0.13–5.08)	1.66 (0.14–20.06)	3.20 (0.69–14.78)	Ref.	1.08 (0.22–5.35)	1.11 (0.16–7.68)
Bisexual Female	1.82 (0.84–3.95)	1.05 (0.27–4.04)	0.76 (0.22–2.69)	1.54 (0.19–12.65)	2.95 (1.37–6.37)	0.92 (0.19–4.56)	Ref.	1.02 (0.24–4.30)
Other Female	1.78 (0.47–6.77)	1.03 (0.18–5.83)	0.75 (0.14–3.92)	1.50 (0.14–16.23)	2.89 (0.75–11.07)	0.90 (0.13–6.26)	0.98 (0.23–4.11)	Ref.

Note. Observation counts (obs.) and sample sizes (n) are unweighted. Adjusted odds ratios (aOR) and confidence intervals (CI) incorporate wave-specific panel analysis weights provided by the Monitoring the Future study. All models control for age (time-varying), survey administration year (time-varying), region, urbanicity, race and ethnicity, parental education, highest degree attained (time-varying), and marital status (time-varying). Sexual identity was treated as time-varying. Only respondents with no lifetime marijuana use at baseline (age 18) were included in the models. Models account for repeated observations with clustering at the respondent-level. Results significant at the 0.05 level are shown in bold. Results for each outcome were adjusted with a Bonferroni correction to maintain a FWER of 0.05 (m = 28).

Table 5
Binary Logistic Regression Results Assessing the Relationship between Sexual Identity and Incident Alcohol and Marijuana Co-Use from Ages 19–30.

	Incident Past 30-Day Alcohol and Marijuana Co-Use (among those who reported no lifetime alcohol use & no lifetime marijuana use at age 18) (obs. = 4461, n = 3498)							
Sexual Identity	aOR (95% CI)	aOR (95% CI)	aOR (95% CI)	aOR (95% CI)	aOR (95% CI)	aOR (95% CI)	aOR (95% CI)	aOR (95% CI)
Heterosexual Male	Ref.	0.31 (0.11–0.85)	0.95 (0.23–3.93)	0.30 (0.05–1.66)	0.86 (0.52–1.43)	0.75 (0.21–2.68)	0.32 (0.13–0.75)	1.88 (0.34–10.38)
Gay Male	3.20 (1.17–8.74)	Ref.	3.04 (0.58–15.79)	0.95 (0.14–6.52)	2.75 (1.01–7.45)	2.41 (0.52–11.20)	1.01 (0.30–3.40)	6.02 (0.88–41.25)
Bisexual Male	1.05 (0.25–4.36)	0.33 (0.06–1.71)	Ref.	0.31 (0.04–2.79)	0.90 (0.22–3.71)	0.79 (0.13–4.98)	0.33 (0.07–1.61)	1.98 (0.23–17.34)
Other Male	3.37 (0.60–18.88)	1.05 (0.15–7.22)	3.20 (0.36–28.50)	Ref.	2.89 (0.51–16.31)	2.54 (0.32–20.14)	1.07 (0.17–6.84)	6.34 (0.60–66.76)
Heterosexual Female	1.16 (0.70–1.94)	0.36 (0.13–0.99)	1.11 (0.27–4.54)	0.35 (0.06–1.95)	Ref.	0.88 (0.25–3.14)	0.37 (0.16–0.87)	2.19 (0.40–12.14)
Lesbian Female	1.33 (0.37–4.73)	0.41 (0.09–1.93)	1.26 (0.20–7.90)	0.39 (0.05–3.13)	1.14 (0.32–4.08)	Ref.	0.42 (0.10–1.79)	2.50 (0.38–16.19)
Bisexual Female	3.16 (1.33–7.50)	0.99 (0.29–3.31)	3.00 (0.62–14.51)	0.94 (0.15–6.02)	2.71 (1.15–6.41)	2.38 (0.56–10.16)	Ref.	5.95 (0.97–36.36)
Other Female	0.53 (0.10–2.93)	0.17 (0.02–1.14)	0.50 (0.06–4.42)	0.16 (0.01–1.66)	0.46 (0.08–2.53)	0.40 (0.06–2.60)	0.17 (0.03–1.03)	Ref.

Note. Observation counts (obs.) and sample sizes (n) are unweighted. Adjusted odds ratios (aOR) and confidence intervals (CI) incorporate wave-specific panel analysis weights provided by the Monitoring the Future study. All models control for age (time-varying), survey administration year (time-varying), region, urbanicity, race and ethnicity, parental education, highest degree attained (time-varying), and marital status (time-varying). Sexual identity was treated as time-varying. Only respondents with no lifetime alcohol use and no lifetime marijuana use at baseline (age 18) were included in the models. Models account for repeated observations with clustering at the respondent-level. Results significant at the 0.05 level are shown in bold. Results for each outcome were adjusted with a Bonferroni correction to maintain a FWER of 0.05 (m = 28).

heterosexual groups, more consistent with minority stress and coping pathways than with distance from substance-involved contexts (Meyer, 2003).

Daily cannabis use is a critical but understudied behavior. Most research focuses on past-month or past-year use (Gonzales, 2020; Schilt-Solberg et al., 2025), yet daily use is especially important because it is linked to cannabis use disorder and mental health concerns (Fischer et al., 2017; Volkow et al., 2014). In our sample, bisexual females had the highest observed prevalence of recent and daily cannabis use, as well as the highest odds of newly-reported daily use, aligning with prior evidence of elevated cannabis involvement among bisexual women (Schuler & Collins, 2020).

Multiple factors may contribute to this disparity. Bisexual women often face "double discrimination" from heterosexual and LGBTQ+ communities, along with bi-erasure and limited support, which minority stress theory links to internalizing symptoms and coping-motivated use (Feinstein et al., 2017; Feinstein & Dyar, 2017; Ford et al., 2023; Meyer, 2003; Taylor, 2018). They also experience higher rates of trauma, including sexual and intimate partner violence, which is strongly associated with cannabis use for coping (Bermea et al., 2018; Coston, 2021; Schuler & Collins, 2020). Cultural factors, including hypersexualization, permissive peer norms, and lower perceived risk, may further elevate use (Johnson & Grove, 2017; Romm et al., 2023; Schulz et al., 2022; Watson et al., 2021), and state-level legalization may widen disparities, as bisexual women report higher cannabis use in states with medical or recreational access (Philbin et al., 2019). More broadly, SM populations experience disproportionately high rates of adverse childhood experiences and other life-course adversity, which are robustly associated with later substance use (Ong et al., 2026; Solberg et al., 2023). Such influences may be especially potent during young adulthood, when transitions in education, employment, and relationships reshape social contexts, potentially explaining why disparities extend to newly-reported use rather than reflecting only earlier use. Because these constructs were not measured in the present study, these explanations are offered as plausible hypotheses rather than established mechanisms.

This study is among the first to examine alcohol-cannabis co-use by both sex and sexual identity in a national young adult sample. Prior work has assessed polysubstance or simultaneous use (Ehlke et al., 2023; Lee et al., 2022), but few studies have measured recent co-use using behaviorally relevant definitions; our past-30-day measure parallels standard definitions of single-substance use and addresses this gap (Bourgault et al., 2022; Patrick, Kloska, et al., 2018). Bisexual females had elevated odds of past-month co-use, more than double those of heterosexual males and females, and gay males also showed elevated odds; incidence analyses showed elevated odds of newly-reported co-use in both groups relative to heterosexual males and females. These patterns may reflect the intersecting psychosocial and structural stressors described above (Johnson & Grove, 2017; Meyer, 2003; Schulz et al., 2022; Watson et al., 2021), and underscore the need for integrated prevention strategies targeting both substances, with particular focus on bisexual females and gay males.

While minority stress theory accounts for elevated risk among sexual minorities relative to heterosexuals, it less readily explains why risk concentrates within specific subgroups. Bisexual males and females may face similar stressors, including biphobia and heterosexism, yet elevated risk concentrated among bisexual females; lesbian females also showed no comparable elevation despite experiencing minority stress. An intersectional framework (Crenshaw, 1989) better accommodates these patterns by recognizing that sexual identity and sex jointly shape exposures, coping resources, and risk, and thus helps explain the uneven distribution of disparities across subgroups. The greater disparities among SM females than males mirror prior NSDUH findings (Schilt-Solberg et al., 2025).

Conversely, lesbian females and bisexual males appeared at lower risk than gay males and bisexual females, with fewer significant elevations. This pattern was not absolute: lesbian females showed elevated incident cannabis use, and bisexual males showed elevated incident daily cannabis use relative to heterosexual females, indicating that risk in these subgroups, though less consistent, is also evident in young adulthood. Potential protective factors include stronger coping skills (Newcomb et al., 2012), social support networks (Frost et al., 2016), growing societal acceptance (Flores, 2021), and gendered resilience factors among lesbian females (Lev, 2015), while bisexual individuals' ability to conceal identity may shape exposure to discrimination differently than among gay males (Balsam et al., 2013). These mechanisms remain hypothesized rather than empirically established in our data; future research should directly test mediators such as social support, coping, identity concealment, and perceived community acceptance.

Race and ethnicity also remain important, given documented racial/ethnic differences in substance use and the predominantly White MTF sample. All models adjusted for race/ethnicity, and unadjusted prevalences in Table 2 mirrored adjusted estimates, suggesting that observed disparities are not solely attributable to racial/ethnic composition. However, sparse cell counts precluded stable interaction estimates, and future work should examine how sexual identity disparities vary across racial/ethnic subgroups.

These findings have direct implications for intervention design. Given risk concentrated among bisexual females and gay males, programming should be tailored rather than uniform across SM subgroups. For bisexual females, trauma-informed care and bi-affirming clinical services that address bi-erasure and double discrimination are warranted (Bermea et al., 2018; Feinstein & Dyar, 2017; Ford et al., 2023), and integrated substance use and mental health services may be particularly important because mental health difficulties are often associated with coping-motivated use among SM populations (Dowling et al., 2022; Ong et al., 2026). For gay males, interventions may benefit from engaging alcohol-centered LGBTQ+ venues where binge drinking norms are reinforced (Adams et al., 2023; Atkinson et al., 2022). Because gay males and bisexual females continue to report new use during young adulthood. Prevalence and incidence patterns also inform intervention timing. Because gay males and bisexual females continue to report new use during young adulthood, this period is an active window for intervention, not only a time when earlier-established disparities persist. For lesbian females and bisexual males, whose elevated use appears more concentrated in patterns established by age 18, earlier prevention may be especially important, although isolated incidence elevations indicate that young-adult services remain warranted.

4.1. Limitations

This study has several limitations. First, MTF is school-based and excludes youth who were not in high school in 12th grade. Because school dropout is associated with both elevated substance use and sexual minority identification, this exclusion likely yields conservative estimates of SM disparities. Second, although the study uses nationally representative longitudinal data, we cannot establish the temporal ordering of identity development and substance use or draw causal conclusions. Relatedly, the mechanisms we invoke were drawn from prior literature and not measured or tested here; they should be treated as hypotheses for future research rather than established explanations. Third, because analyses were stratified by binary sex, with other groups excluded, gender minority young adults were not represented in primary comparisons. Gender minority individuals experience distinct minority stressors, and their exclusion limits what this study can say about SM young adult substance use as a whole; future research should prioritize these populations using survey designs that adequately power such analyses. Fourth, survey year was not significantly associated with the outcomes. Because year was included as a covariate rather than examined through stratified models, however, these analyses adjust for period effects rather than test whether identity-based disparities varied across the pandemic period. Finally, findings reflect the U.S. context, where

cannabis legalization, LGBTQ+ rights frameworks, and healthcare access differ from other countries, and may not generalize internationally.

5. Conclusion

In summary, young adult gay males and bisexual females engaged in elevated substance use relative to heterosexual young adults, with differences in prevalence, odds, and incidence, while lesbian females and bisexual males showed comparatively lower risk. These findings highlight the need for targeted, culturally competent interventions and for research that examines underlying mechanisms and follows SM young adults prospectively. Addressing these disparities is essential to promoting health and well-being across the broader SM community.

CRediT authorship contribution statement

Marvin A. Schilt-Solberg: Writing – review & editing, Writing – original draft, Funding acquisition, Data curation, Conceptualization. **Philip T. Veliz:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation. **John Jardine:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation. **Yuk C. Pang:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation. **Sean Esteban McCabe:** Writing – review & editing, Writing – original draft. **Rebecca Evans-Polce:** Writing – review & editing, Writing – original draft. **Megan E. Patrick:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Funding acquisition, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.abrep.2026.100732.

Data availability

Data will be made available on request.

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